SFUIM: A Four-Factor Adaptive User Interaction Model for Dynamic Prompt Generation in LLM Systems, Derived from the FFUIM Framework

Participant Information Sheet (PIS)

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The information in this section is to help you decide whether or not you want to participate in this research study by providing a summary of key points about the study. We have included detailed information after this section.

- **What is the study about:**
  This study investigates how personalised prompt adaptation can improve user understanding and satisfaction when interacting with an AI-based question answering system.
- **What participation involves:**
  You will interact with an online question answering system, provide short ratings during the interaction, and complete a brief questionnaire at the end.
- **Who is running the study:**
  The study is conducted by a student researcher at the University of Southampton.
- **What happens to the data:**
  All data will be collected anonymously and analysed in aggregated form.
- **Payment:**
  You will receive a £5 compensation for your time.

- We invite you to take part in our research study. Before you decide, please take time to read this information. It explains why we are doing the study and what it will involve.
- If anything is unclear or if you have any questions, please ask.
- You can talk to someone you trust before you decide. It's entirely your choice whether to take part.
- If you decide to take part, you will be asked to sign a consent form.

What is the research about?

My name is **Xinlu Dong**, and I am a student at the University of Southampton studying towards an integrated Bachelor's and Master's degree in Computer Science and Artificial Intelligence. This study is part of my individual research project.

The system is based on a Large Language Model (LLM), which is an artificial intelligence model capable of generating text responses to user questions. The aim of this research is to evaluate an adaptive interaction framework (SFUIM) that personalises how an AI

system responds to users based on feedback. The study compares a personalised system with a non-personalised or partially personalised version to understand whether adaptive prompts improve clarity, usefulness, and learning experience.

This research has been reviewed and approved by the University of Southampton's Faculty Ethics Committee.

Why have I been asked to take part?

You are being invited to take part because you are aged 18 or above and have basic experience using online question answering or conversational systems.

Approximately 30**–35 participants** will take part in this study. Participation is entirely voluntary.

What will happen to me if I take part?

If you take part, you will:

- *Receive a survey link to access and complete the study online via a web-based interface.*
- *Interact with four versions of an AI-based question answering system.*
- *For each system, engage in a multi-round dialogue of up to 10 interaction turns. The system will automatically indicate when the maximum number of turns has been reached.*
- *After each system response, provide brief ratings (for example, clarity and usefulness).*
- *Move to the next system once you feel you have sufficiently understood the topic or when the maximum number of turns is reached.*
- *Complete a short post-study questionnaire after finishing all four systems, reflecting on your experience and understanding.*

You will interact with four versions of the system during the study:

- a full personalised version (SFUIM),
- a baseline version (non-personalised),
- a version without the time factor,
- and a version without the frequency factor.

You will not be informed which version you are using at any given time.

The study will take approximately 40–60 minutes in total. The study is conducted entirely online. No audio or video recording will be used. The researcher will not be physically present during participation.

Are there any benefits in my taking part?

You may not receive a direct personal benefit. However, your participation will help improve understanding of how AI systems can better support users' learning and information needs.

You will receive **£5 compensation** as a token of appreciation for your time. This compensation is not dependent on performance or completion and will still be provided if you choose to withdraw early.

Are there any risks involved?

This study is considered low risk. There are no known physical or psychological risks. You may experience minor inconvenience or fatigue from using the system or answering questions. You are free to stop at any time.

What data will be collected?

The study will collect:

- Basic demographic information (age, gender, education level, field of study)
- Ratings and feedback provided during system interaction
- Responses to a short post-study questionnaire

No names, contact details, or identifying information will be collected.

How will my information and personal details be protected?

- Your participation and the information we collect about you during the research will be kept strictly confidential during the study.
- Only the research team and approved staff from the University of Southampton may see your data to help monitor the study. They may also check the study to make sure it complies with research regulations.
- People from regulatory organisations (who make sure the study follows proper rules) might also need access to your data.
- Everyone who sees your information is required to keep it confidential.

All data will be collected anonymously and stored securely on password-protected devices and university-approved storage. Only the researcher and supervisor will have access to the data.

As this is an undergraduate/master's project, all research data will be securely deleted after the degree is awarded.

Do I have to take part?

- No, it is entirely up to you to decide whether to take part. If you decide to take part, you will need to indicate consent to show you have agreed to take part.
- If you decide to take part,you can contact my school email. You will use the test link to participate this study.

What happens if I change my mind?

- You have the right to change your mind and are free to withdraw at any time without giving a reason
- Because the study is anonymous, it will not be possible to remove your data after submission.

What will happen to the results of the research?

- Reports and publications will not directly identify you without your specific consent.
- The results will be analysed and reported in anonymised and aggregated form as part of a university research project.No individual participant will be identified.

Where can I get more information?

If you have questions, please contact:

Xinlu Dong – xd3g23@soton.ac.uk

What happens if there is a problem?

- If you are concerned or unhappy about any part of the study, please contact the researchers. They will do their best to answer your questions.
- If you remain unhappy or have a complaint about any part of this study, please contact the Head of Research Ethics and Governance (023 8059 5058, rgoinfo@soton.ac.uk), University of Southampton.

Data Protection Privacy Notice

This Participant Information Sheet tells you what data will be collected for this project and whether this includes any personal data. Please ask the research team if you have any questions or are unclear what data is being collected about you.

How will we use information about you?

Any personal data we collect in this study will be used only for the purposes of carrying out our research. Contact details *will not* be retained beyond the end of the study or used for future studies. Your personal data will be handled according to the University's relevant policies.

The University of Southampton is the 'Data Controller' for the personal information we collect from *you* for this research project.

Information that may identify you, such as age, gender, education level, and field of study, will be used to carry out the research or to check your records to make sure that the research is being done properly.

We will keep all information about you safe and secure.

Once the study is finished, your identifiable information will be deleted.

We will write our reports in a way that no-one can work out that you took part in the study.

To safeguard your rights, we will use the minimum personal data necessary to achieve our research study objectives. Your data protection rights – such as to access, change, or transfer such information - may be limited, however, in order for the research output to be reliable and accurate. The University will not do anything with your personal data that you would not reasonably expect.

We will keep all information about you safe and secure.

What are your choices about how your information is used?

- You can stop being part of the study at any time, without giving a reason, but we will keep all *anonymous* information about you that we already have.
- We need to manage your records in specific ways for the research to be reliable. This means that we won't be able to let you see or change the data we hold about you.

Where can you find out more about how your information is used?

You can find out more about how we use your information:

- by sending an email to University's Data Protection Officer (data.protection@soton.ac.uk).
- by asking one of the research team or from our general privacy policy.
- by sending an email to [xd3g23@soton.ac.uk], or by ringing us on [+44 7786 925695].

How long we keep your data?

As this is a taught undergraduate/postgraduate student project, research data will not be retained after the degree has been awarded and will be securely deleted once the project has been completed.

Thank you.

Thank you for taking the time to read the information sheet and considering taking part in the research.